

Rachel Ratajczak

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Healthcare Data Analyst

Highly analytical M.S. in Computer Science graduate with a B.S. in Integrative Biology, offering a unique blend of advanced technical expertise and clinical laboratory experience. Skilled in extracting, cleaning, validating, and interpreting complex datasets to deliver actionable, data-driven insights. Proven track record of managing structured clinical databases, ensuring data integrity, and following standardized reporting requirements. Adept at collaborating cross-functionally with clinicians, scientists, and technical stakeholders to support operational decision-making and patient outcomes.

CORE SKILLS

Data Analytics & Programming: Python (pandas, seaborn, matplotlib), R (tidyverse), SQL, Advanced Excel, Power BI, Tableau, Machine Learning, Data cleaning, ETL

Clinical & Healthcare: Molecular research, HIPAA compliance, Laboratory Information Systems (LIS), EHR systems, Clinical documentation, Clinical workflow, Patient outcomes

Data Management & QA: Data preparation, relational database systems, missing value imputation, Quality Assurance (QA)

EDUCATION & CERTIFICATIONS

Master of Science (M.S.) in Computer Science, University of Illinois at Springfield (UIS) | 2025

Bachelor of Science (B.S.) in Integrative Biology, Michigan State University (MSU) | 2020

Google Data Analytics Professional Certificate, Coursera | In Progress

KEY PROJECTS

CDC Diabetes Health Indicators – Exploratory Data Analysis (Python)

- Analyzed the CDC BRFSS 2015 healthcare dataset to identify behavioral risk factors associated with diabetes.
- Utilized Python (pandas, seaborn, matplotlib) to clean data, analyzed feature correlations, and generated visual distributions for BMI, age trends, and blood pressure risk factors.

PCOS Diagnosis Classification & Prediction (R)

- Developed a machine learning classification model using R to predict Polycystic Ovary Syndrome (PCOS) based on patient risk factors.
- Performed rigorous data cleaning, feature evaluation, and statistical visualization of clinical variables.

WORK EXPERIENCE

Certified Nursing Assistant | Assisting Hands Home Care – Chicago, IL

2026

- Accurately recorded and documented daily vital health metrics and patient observations within digital health tracking and EHR systems, maintaining strict data integrity for clinical care teams.
- Partnered with multidisciplinary care providers to track patient progress, reporting emerging clinical trends and changing health conditions to support informed, data-driven care decisions.

Medical Laboratory Technologist | Gago Center for Fertility – Brighton, MI

2022 - 2023

- Managed high-stakes fertility data in Excel-based databases, implementing strict tracking protocols to maintain 100% data integrity and eliminate sample mismatches for critical IVF/IUI treatments.
- Reported complex and time-sensitive clinical specimen and blood panel data into Electronic Health Record (EHR) systems, maintaining high data accuracy and streamlining the decision-making pipeline for multidisciplinary medical teams.

Molecular Technologist | Great Lakes Medical Laboratory – Farmington Hills, MI

2020 - 2022

- Processed hundreds of COVID-19 samples daily, executing strict quality control protocols to ensure maximum data accuracy and zero contamination in high-throughput laboratory operations.
- Co-developed alternative sample processing methodologies using saliva collection for elderly demographics, optimizing data-gathering workflows to improve patient accessibility and processing efficiency.

Research & Development Scientist | Archer Daniels Midland – Decatur

2020

- Synthesized complex experimental data of specialized lecithin properties into comprehensive reports and data visualizations, presenting key findings during weekly team meetings to drive collaborative R&D decision-making.
- Mastered numerous analytical instruments to evaluate structural data trends of lecithin products, authoring Standard Operating Procedures (SOPs) to ensure consistent data collection methods across the laboratory.